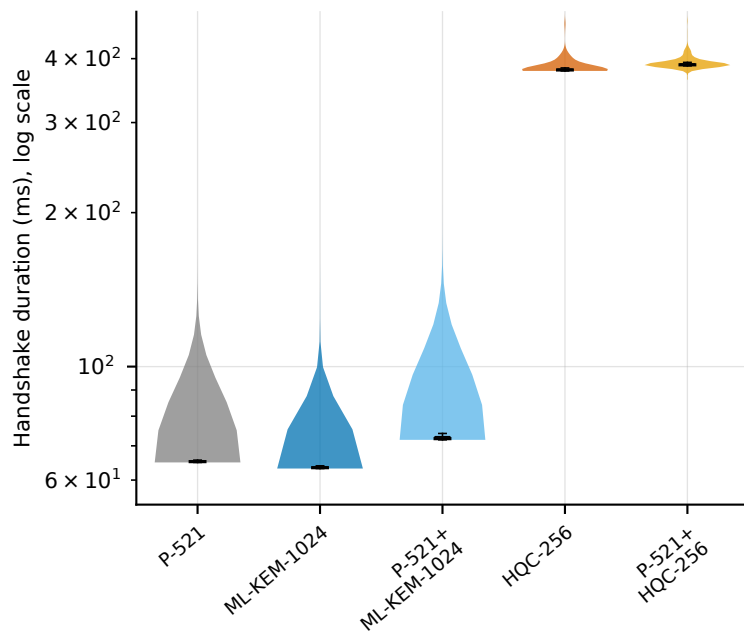
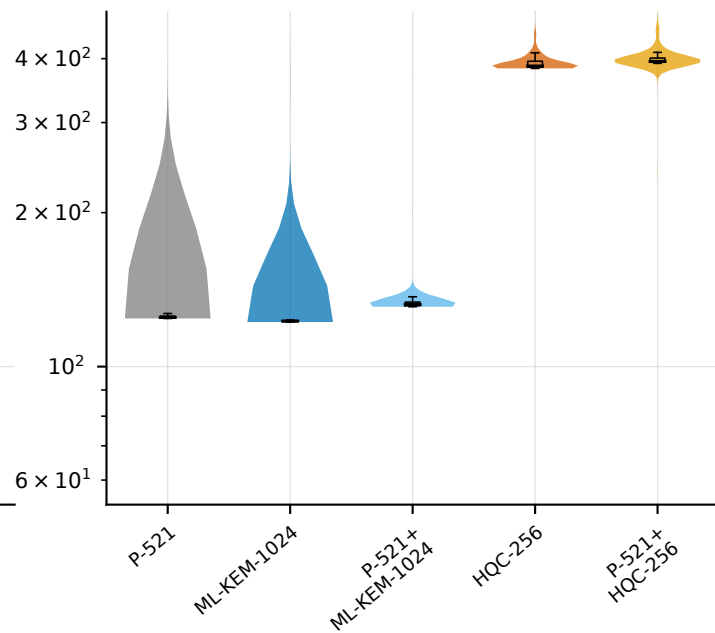


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	71.7	107.8	1.06k	143.4	163.0	3.19k
ML-KEM-1024	66.3	81.5	1.27k	131.9	101.9	2.24k
P-521+ML-KEM-1024	80.7	115.6	1.28k	133.8	8.6	375.0
HQC-256	386.4	7.1	704.2	393.4	6.3	854.5
P-521+HQC-256	393.3	4.4	712.8	402.2	8.3	862.9